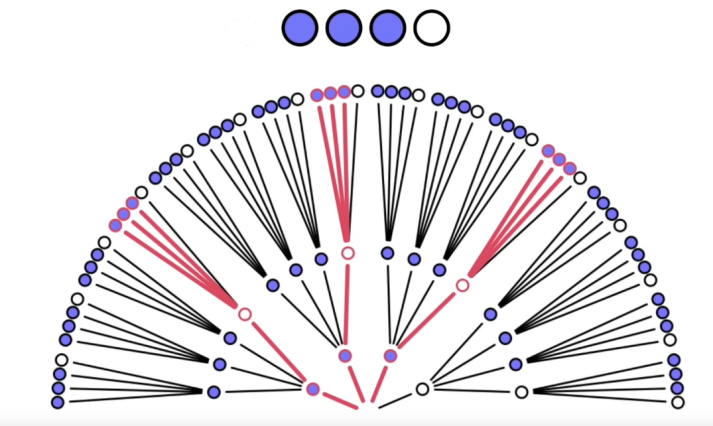
# Lecture 6: Chapter 2 - Small Worlds and Large Worlds

- Small world: self-contained logical world of the model: all outcomes are identified and quantified from a plausibility perspective. No surprise. Should perform as expected under assumed conditions.
  - Large world: larger context of the real world where the model is deployed. Events we didn't think of can occur. The model is an incomplete representation of the world.
- Bayesian models learn optimally in the small world context. Large model performance has to be shown empirically as it cannot really be deduced.
- We focus here on the small world and show that probability is essentially a counting exercise (sometimes continuous counting so integrating). Bayesian inference stems naturally from this perspective.

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# Garden of forking data

- It's all about counting in the world where we count everything that could have happened and compare it to options compatible with our data



- We have conjectures (how many balls in the bag) and to figure out the plausibility, we gather evidence about the content of the bag (our observation(s))

| Possible composition | $p$ | ways to produce data | plausibility |
|---|---|---|---|
| [○○○○] | 0 | 0 | 0 |
| [●○○○] | 0.25 | 3 | 0.15 |
| [●●○○] | 0.5 | 8 | 0.40 |
| [●●●○] | 0.75 | 9 | 0.45 |
| [●●●●] | 1 | 0 | 0 |

```
ways <- c( 3 , 8 , 9 )
ways/sum(ways)
```

```
[1] 0.15 0.40 0.45
```

# Updating

- We can handle all the data at once or take all the observation up to a point as a prior and then update our belief given a new observation into a posterior.
- Both methods are equivalent. Multiplication is a shortcut for counting.

Another draw from the bag: ●

| Conjecture | Ways to produce ● | Previous counts | New count |
|---|---|---|---|
| [○○○○] | 0 | 0 | $0 \times 0 = 0$ |
| [●○○○] | 1 | 3 | $3 \times 1 = 3$ |
| [●●○○] | 2 | 8 | $8 \times 2 = 16$ |
| [●●●○] | 3 | 9 | $9 \times 3 = 27$ |
| [●●●●] | 4 | 0 | $0 \times 4 = 0$ |

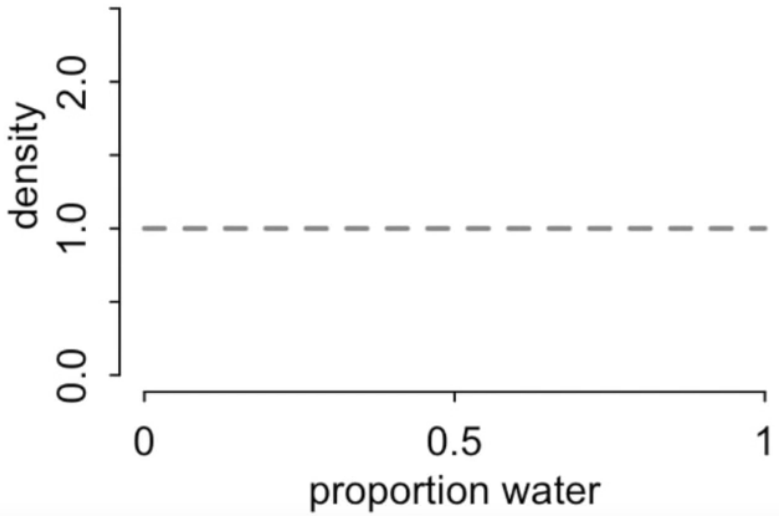
- Rules for Bayesian updating:
- (i) State a causal model for data generation, given each possible explanation, (ii) count ways the data could arise for each explanation and (iii) relative plausibility is relative counts from (ii).

# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?



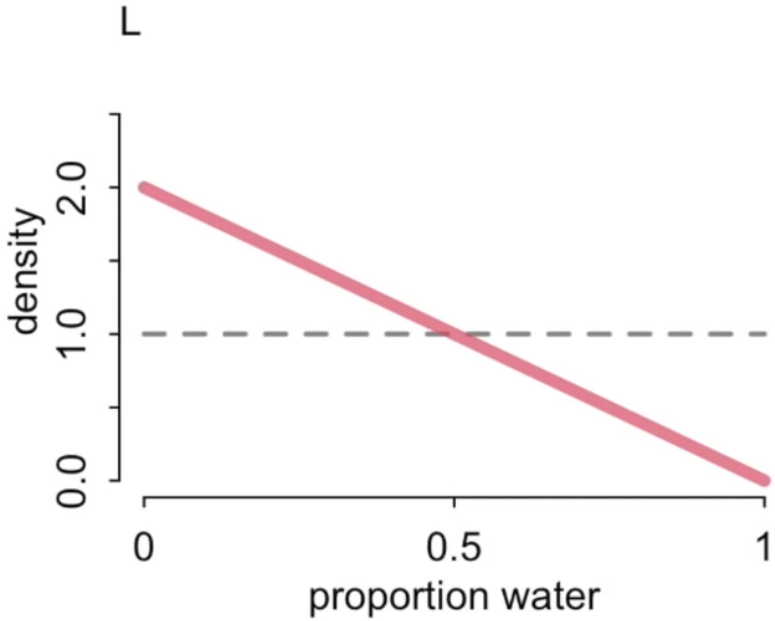


# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?



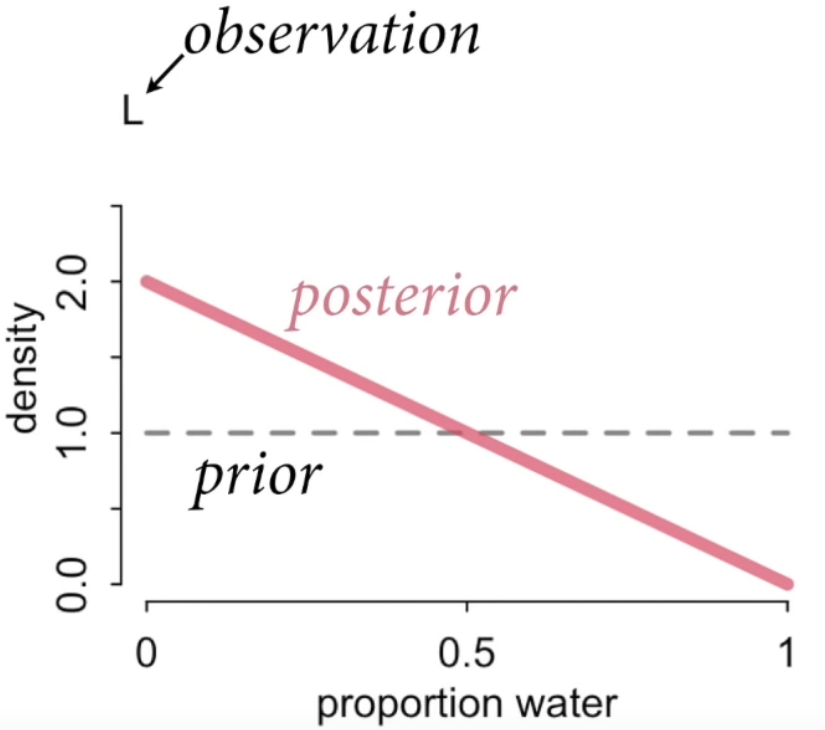



# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?

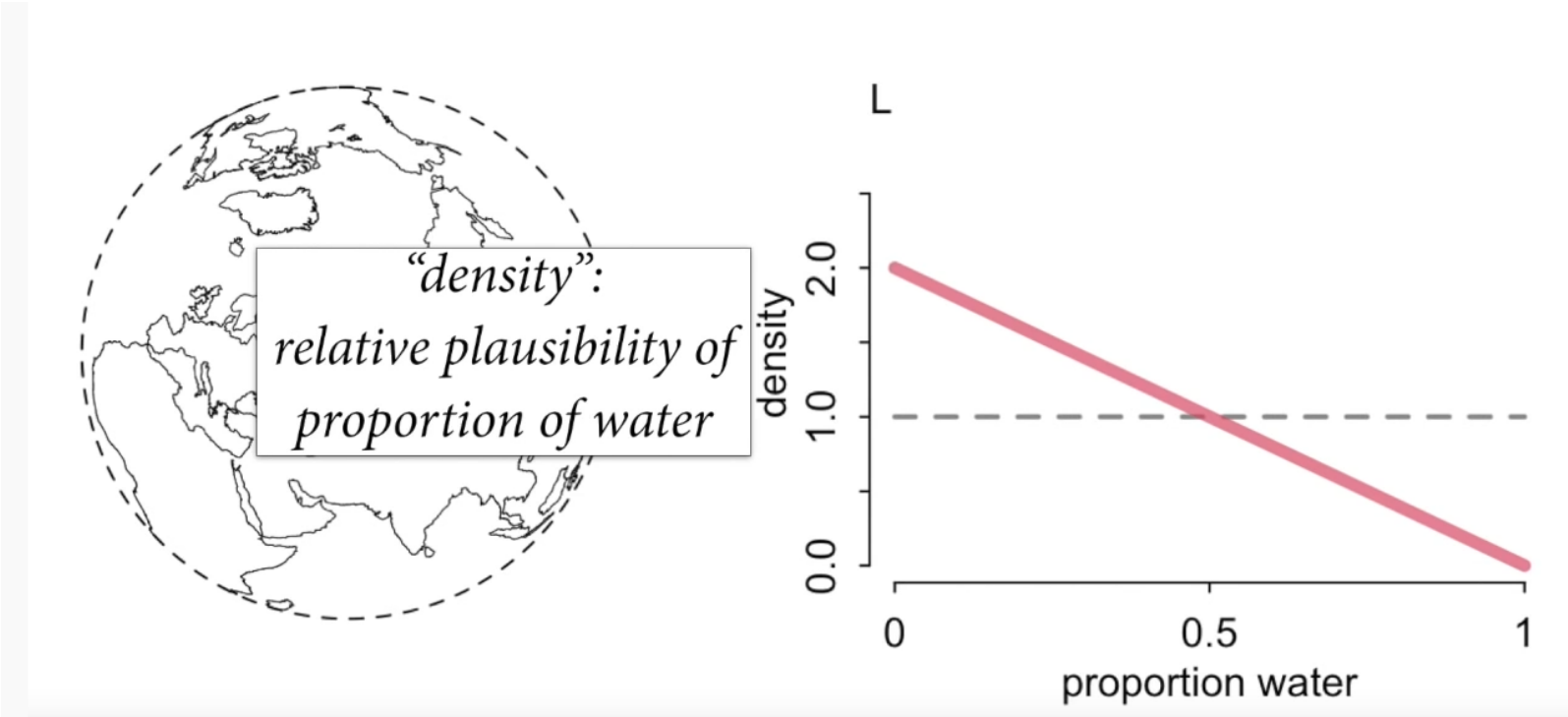


# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?

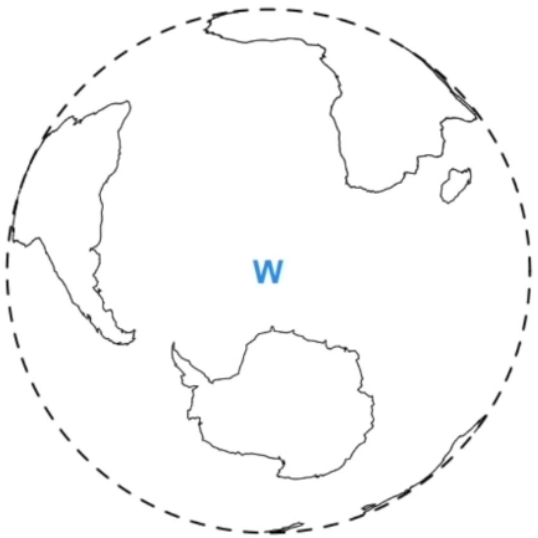


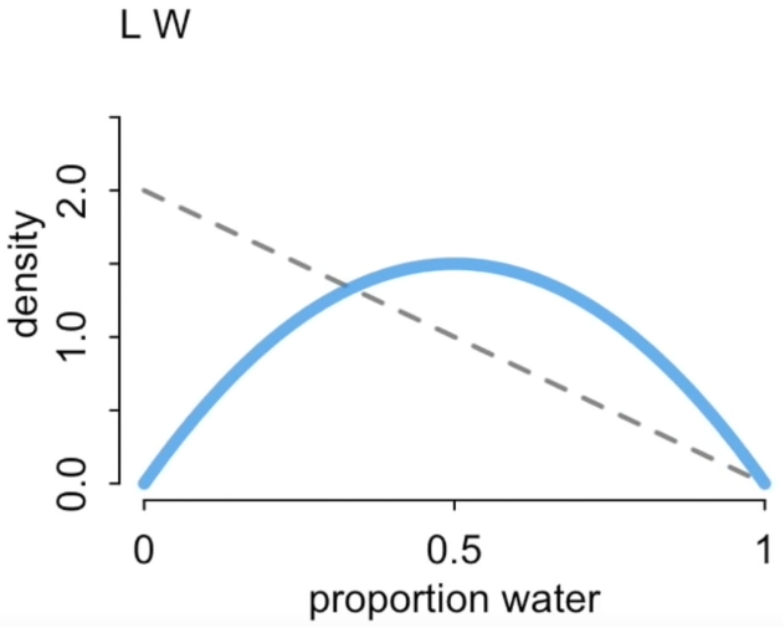

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# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?

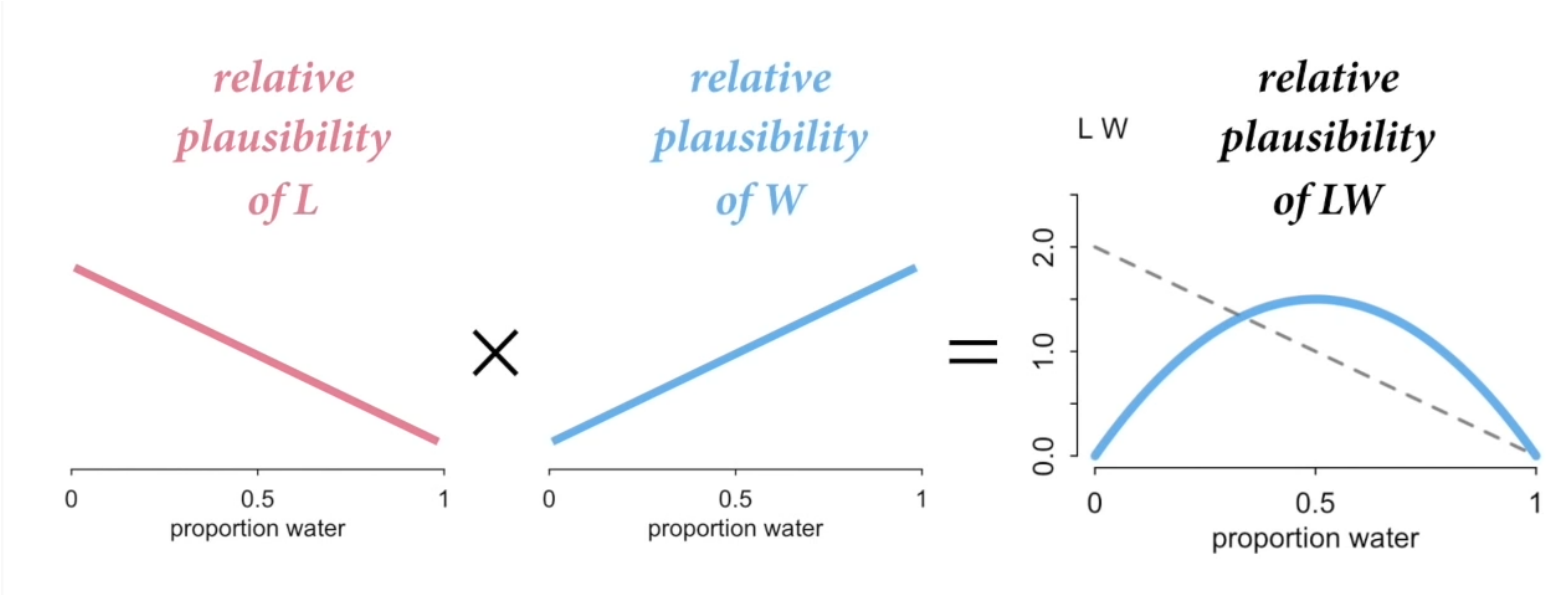


# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?

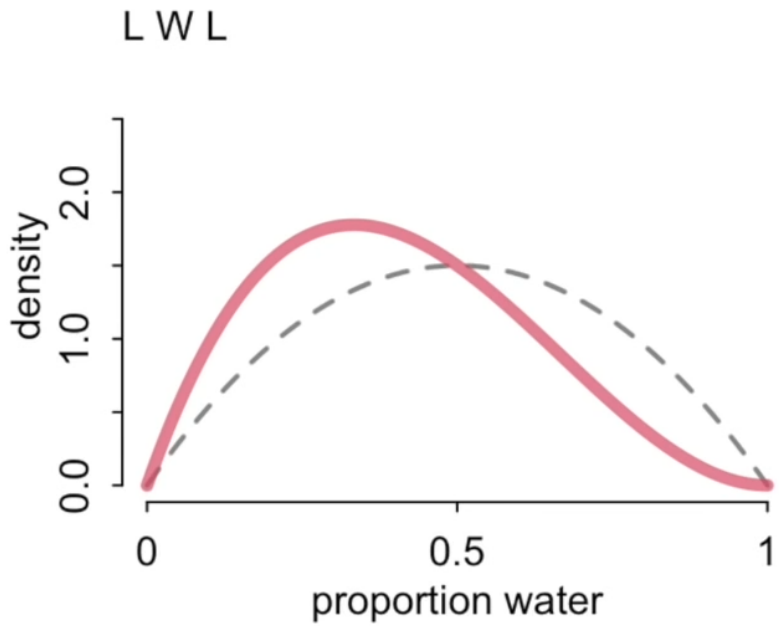


# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?



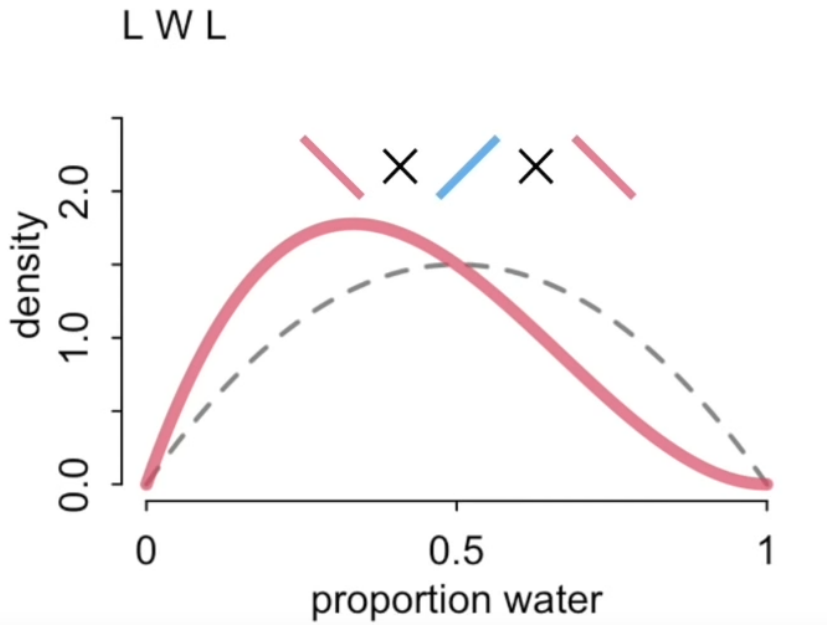



# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?

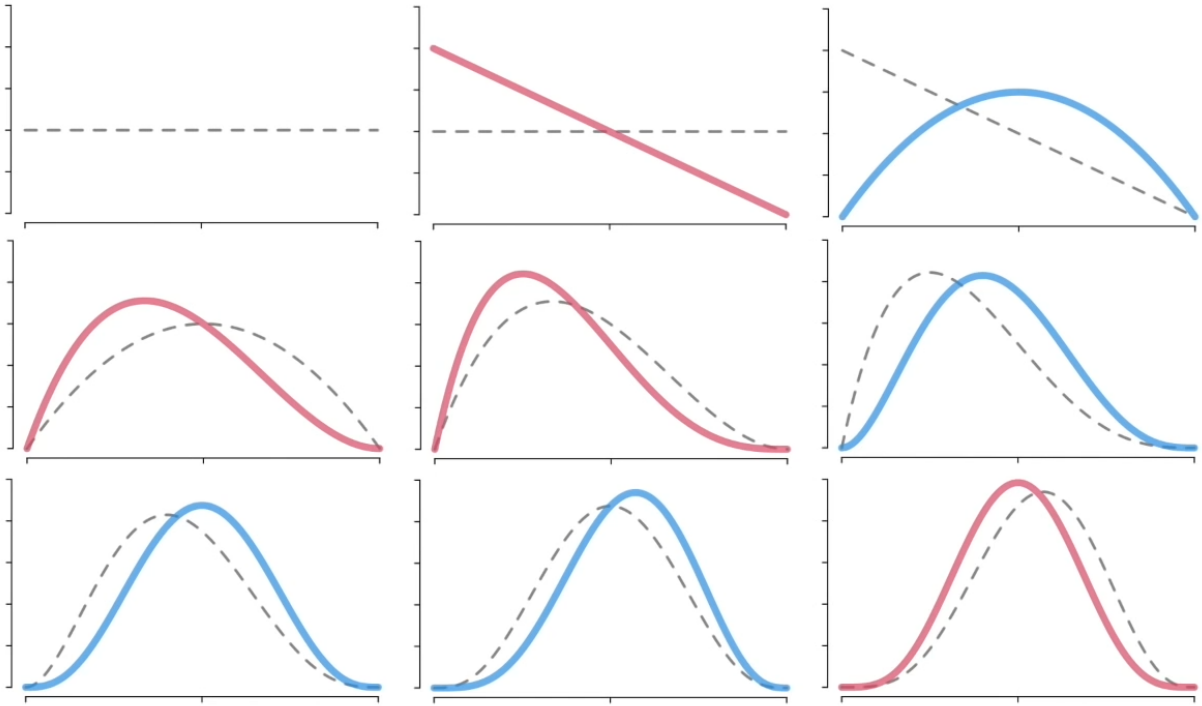


# Globe of forking water

## Continuous infinite number of possible explanations

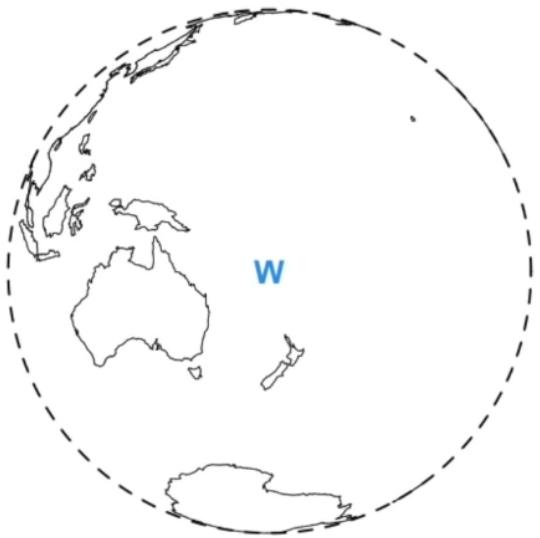
How much water on the surface of Earth?

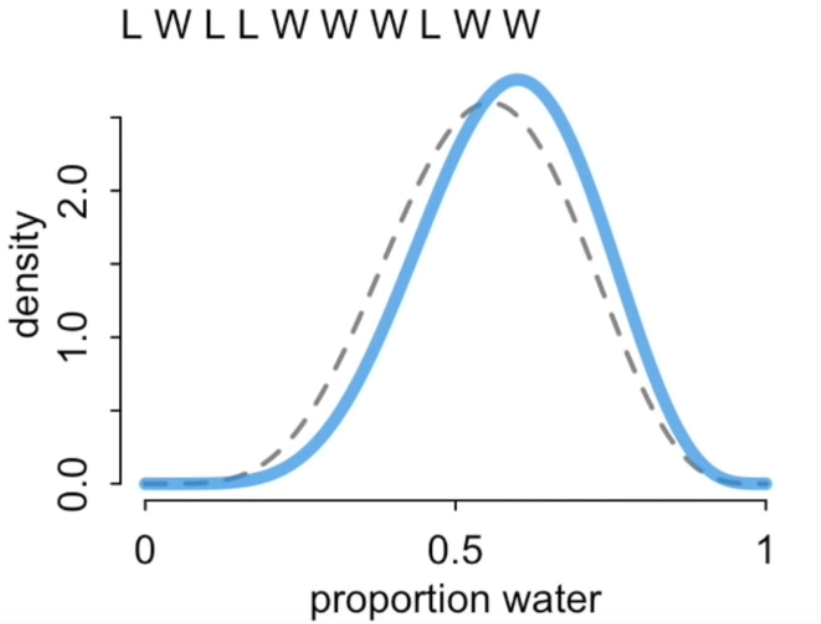


# Globe of forking water

## Continuous infinite number of possible explanations

How much water on the surface of Earth?



# A few ingredients to build our plausibilities

- The formula (Bayes' theorem, right?)

$$\text{Plausibility(parameter)}_{\textit{after } D_{new}} = \frac{\#(\text{Ways})_{\text{parameter } \textit{can produce } D_{new}} \times \text{Prior plausibility(parameter)}}{\textit{sum of products}}$$

- Bayesian model design loop is logic
  1. Data story: motivate the model by narrating how the data might arise. Can be descriptive (association to predict outcome given observations) or causal (theory of events producing other events).
  2. Update: Educate the model by feeding it data.
  3. Evaluate: all statistical models require supervision, possibly leading to model revision(s). Check model's inferences in light of aspect of the data it (should) ignore. Simulations can help.

# A few ingredients to build our plausibilities

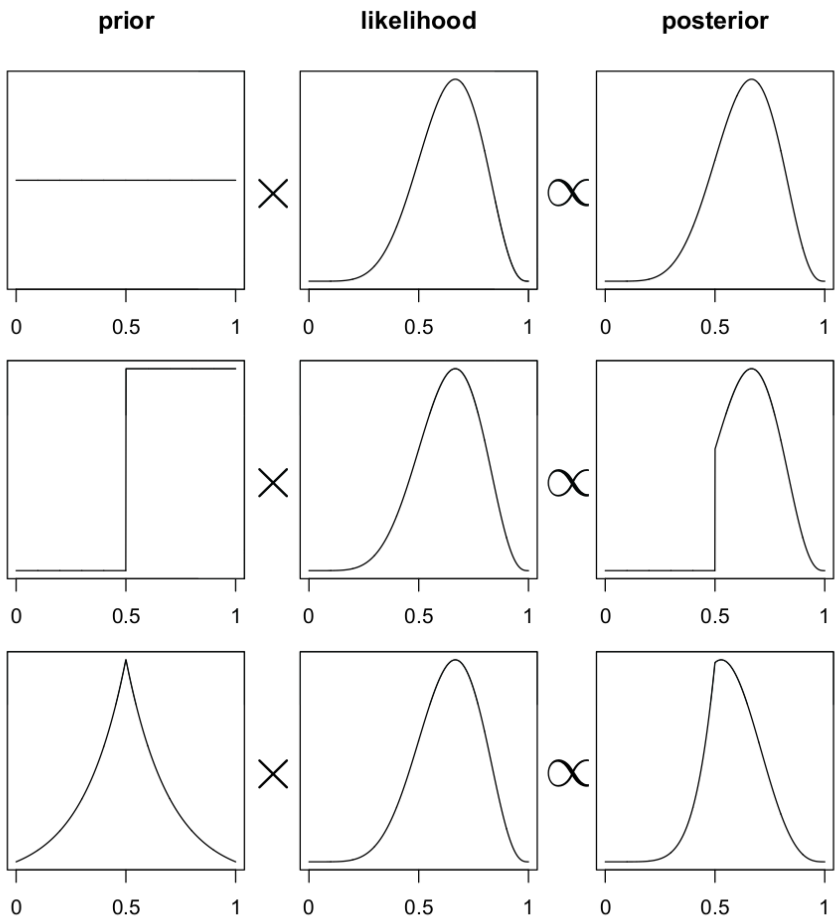
- A *parameter* is a conjectured explanation in the model. There can be more than one.
- The relative number of ways a parameter value can produce the data is called the *likelihood* of that parameter value. We enumerate all possible data sequences that could have happened and eliminate inconsistent sequences given the evidence (the data).
- The *prior probability* is the prior plausibility on the parameter. It can be derived form "old" data or from external expertise.
- Last but not least, the new updated plausibility after seeing the new data $D_{new}$ is called the *posterior probability*.

# A first discussion on priors

- Where do the prior come from?
- Both engineering assumptions (to help/allow the machine to learn) and scientific assumptions (to reflect our knowledge on the phenomenon).
- *Subjective Bayesian* advocate for a choice of priors based on personal beliefs of the analyst.
- A flat prior is not uninformative in many situations. And possibly not always the most clever choice either.
- Any prior must be chosen, evaluated and revised just like any component of the model.
- In the absence of strong argument in favour of a prior, try different ones $\rightarrow$ *sensitivity analysis* of the conducted inference to the priors/assumptions.
- Yes we can "lie" with priors, but just like with any model component (including the likelihood). Not very clever as it is highly visible!

# Multiplicative interaction of prior and likelihood

## To produce the posterior



# A few characteristics of our Bayesian updating

1. No minimum sample size. Some implication of the model can be understood without data (see prior predictive simulations). Ideally, more data is better though.
2. Shape embodies sample size: curve narrower and taller as sample size increases.
3. No point estimate. *The estimate is the distribution/curve*. You can look at summaries of the curve as a last step to avoid taking away uncertainty: mean, median, mode...A good choice depends on what you want to show.
4. Similarly, no true interval. They can help communicate the shape of the posterior and/or ranges of plausible values. Note that nothing happens at the boundary, especially the 95% conventional one.

# How to write a model/draw the owl

- We write the model in a way that communicates assumptions about the probabilities.
  Data: $W$ and $L$ the number of water and land observations, respectively

$$P(W, L|p) = \frac{(W+L)!}{W!L!} p^W (1-p)^L$$

Binomial probability function

```
dbinom(W, W+L, p)
```

(try values for $W$, $W+L$ and $p$ in R)
We also need to specify a prior for $p$, let's assume a constant, i.e. maximum ignorance:

$$P(p) = 1/(1-0) = 1$$

as we must say how each variable is generated.

# How to write a model/draw the owl

- We write the model in a way that communicates assumptions about the probabilities.
- Posterior is the normalised product:

$$P(p|W,L) = \frac{P(W,L|p)P(p)}{P(W,L)}$$

(multiplying is like counting in the garden; can ignore $P(W,L)$ in most cases)

- Let's do it computationally

# Engine of the (Bayesian) machine

## As many ways to count

You need a motor to process data with likelihood, combine with priors and produces a posterior $\approx$ conditioning the prior on the data. Some priors lead to "easier" (or closed-form) calculations. We don't want forced model choices for technical reasons.

- Grid approximation (can be quite inefficient if #(parameters) grows)
- Other methods
  - Quadratic (or Gaussian) approximation relies on stronger assumptions (soon): $\log(\text{Posterior}) \sim$ Parabola
  - MCMC (later)

# Grid approximation

## Not always efficient but good for concept

```
grid.resolution <- 3 # or 5, 10, 20, 1000
p_grid <- seq(from=0, to=1, length.out = grid.resolution)
prob_p <- rep(1, grid.resolution)
prob_data <- dbinom(W, size = W+L, prob = p_grid)
posterior <- prob_data*prob_p
posterior <- posterior/sum(posterior)
```

Line 1 sets up the grid, aka the possible explanations
Line 2 sets up the prior probability for each grid value
Line 3 give the likelihood, i.e. the probability of the data for all (grid) explanations
Line 4 compute the relative posterior and Line 5 normalises the posterior.

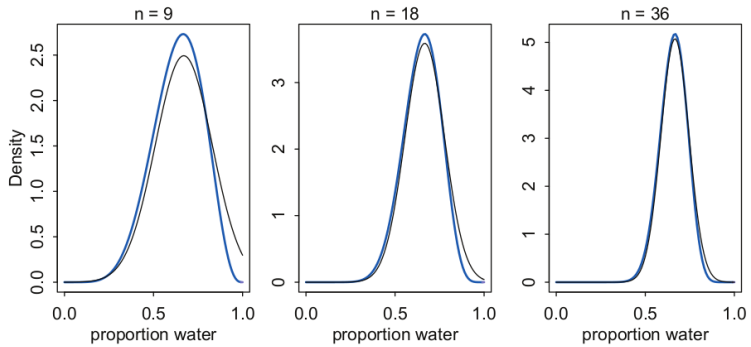
Change the prior, e.g. to `prob_p <- dbeta(p_grid, 3,1)` or to `prob_p <- ifelse(p_grid<0.5, 0, 1)`

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# Quadratic approximation

## Often OK when *n* becomes moderately large (see SR)

```
library(rethinking)
globe.ca <- quap(
alist(
W ~ dbinom(W+L,p) # binomial likelihood
p ~ dunif(0,1) # uniform prior
),
data=list(W=6,L=3)
)
precis(globe.qa) # displays summary of approximation
```




Read Section 2.4.4 in details in SR.

# Markov Chain Monte Carlo

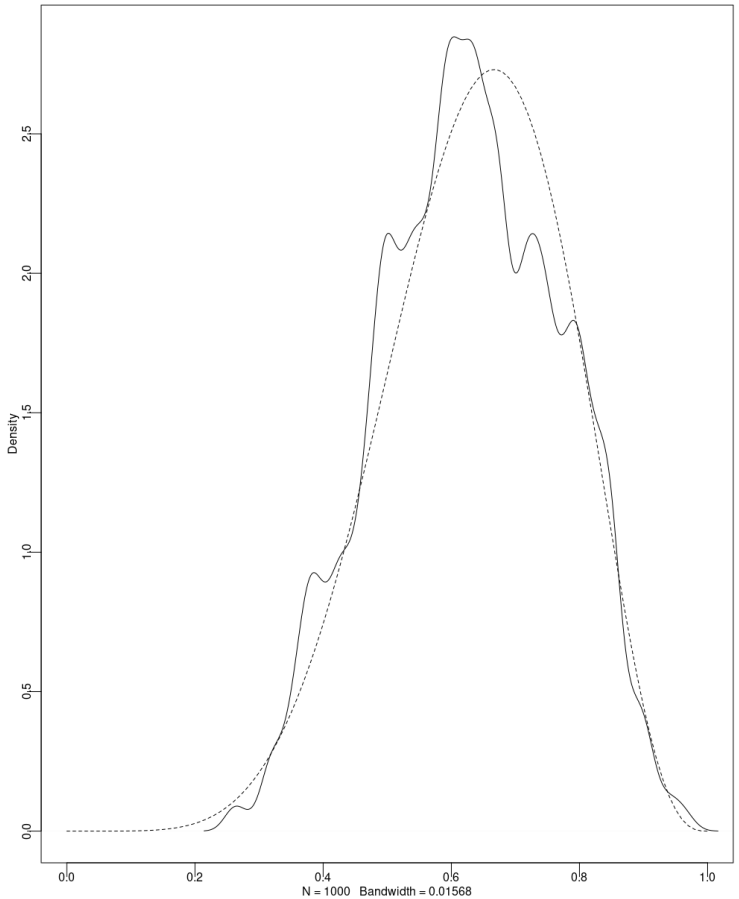
## Metropolis algorithm (more later)

Can handle highly complex models. Does not attack the posterior approximation directly, but draws samples form it.

```
n_samples <- 1000 # number of samples drawn
p <- rep(NA, n_samples)
p[1] <- 0.5 # initial value
W <- 6, L <- 3 # data
for(i in 2:n_samples){
p_new <- rnorm(1, p[i-1], 0.1) # proposed updated parameter
if (p_new < 0) p_new <- abs(p_new) # mirror conditions
if (p_new > 1) p_new <- 2 - p_new # at the (0,1) boundaries
q0 <- dbinom(W, W+L, p[i-1]) # old likelihood
q1 <- dbinom(W, W+L, p_new) # new likelihood
# acceptance ratio
p[i] <- ifelse(runif(1) < q1/q0, p_new, p[i-1])
}
dens(p, xlim = c(0,1))
curve(dbeta(x, W+1, L+1), lty = 2, add = TRUE)
```

# MCMC

## Posterior approximation for the Land vs Water proportion example



# From posterior to predictions

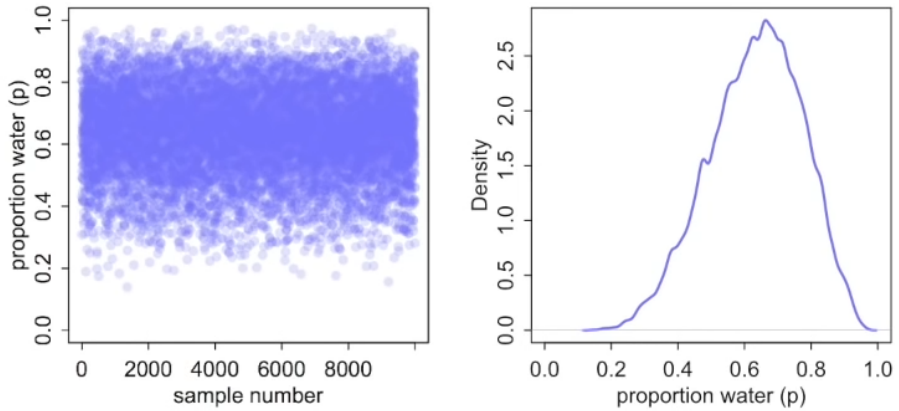
- Model implications depend on the entire posterior
- The posterior can be for many many parameters. And their interactions.
- We must average any inference of the entire posterior.
- Mathematically, this means summing or integrating over a posterior.
- We can just take samples form the posterior and do calculus on those to study implications.
- $\rightarrow$ Posterior predictive distributions: model-based forecasts, causal effects, counterfactuals. Can also simulate from priors.

# From posterior to predictor

## Video of predictions

A video to go from posterior uncertainty (left-hand side distribution on $p$), to a predictive distribution of number of $W$ out of 9 samples via a causal model, to the implications for the number of waters over all possible $p$ drawn according to the posterior (right-hand side distribution). The right-hand side is called the *posterior predictive distribution*.

```
samples <- sample(p_grid, prob = posterior, size = 10000,
                  replace = T)
```




(see next week, Chapter 3, SR)

# Summary

- Introduced the concept of the Bayesian machinery.
- The goal is to estimate the posterior distribution of the parameters.
- Obtained via counting the relative number of ways each data explanation (parameter) can cause the data $\rightarrow$ indicates plausibilities in the different explanations.
  *Explanations with more ways to produce the data are more plausible*
- Different technical components: (i) prior of the parameter (before data), (ii) likelihood of the data for a value of the parameter and (iii) use of Bayes' theorem to combine them into the posterior.
- In practice, we have numerical techniques to estimate the posterior: grid approximation, quadratic approximation or MCMC.
- We can have a look at the implications in terms of predictions, e.g. for new data from the causal model.